

HPV vaccination simulation model

Cost-benefit analysis of extending vaccination age cap within VHA
Updated slides based on **new transition model**.

Needs more work!! **Calibration of parameters for VA**

2025-04-21

Agent-based models

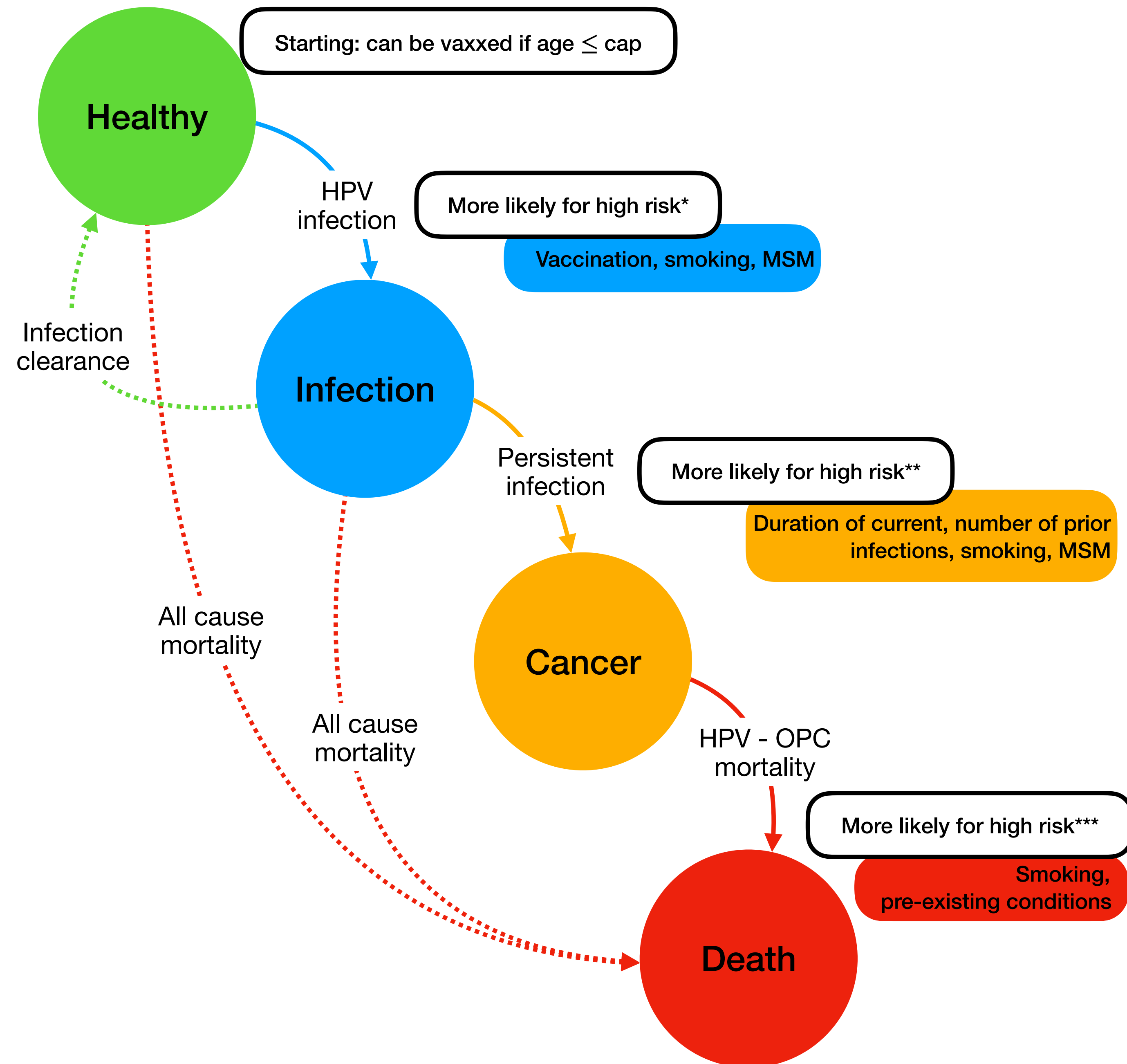
- Step 1: Simulate heterogenous population of agents
 - **Agents:** with features.
 - **Men in VA:** tobacco use + vaccine hesitancy + age
- Step 2: Model actions and interactions with environment
 - **Environment and Interactions:** MSM + vaccine hesitancy.
- Step 3: Flexible tracking of trajectory over time.
 - **Behavior changes:** vaccine uptake + smoking + MSM.

Designing an ABM Transition diagram

Attributes of agent

1. **20 ≤ Age ≤ 65**: will simulate according to VA.
2. **Vaccination status: Y/N**: will simulate according to VA.
3. **HPV status: Positive/Negative**: will need input.
4. **Duration of infection (in years): ≥ 0**
5. **Number of previous infections: ≥ 0**
6. **Covariates**: can change over time
 1. *Vaccine hesitancy*
 2. *Smoking history*
 3. *# of sexual partners*

Primary pathway of interest: solid lines

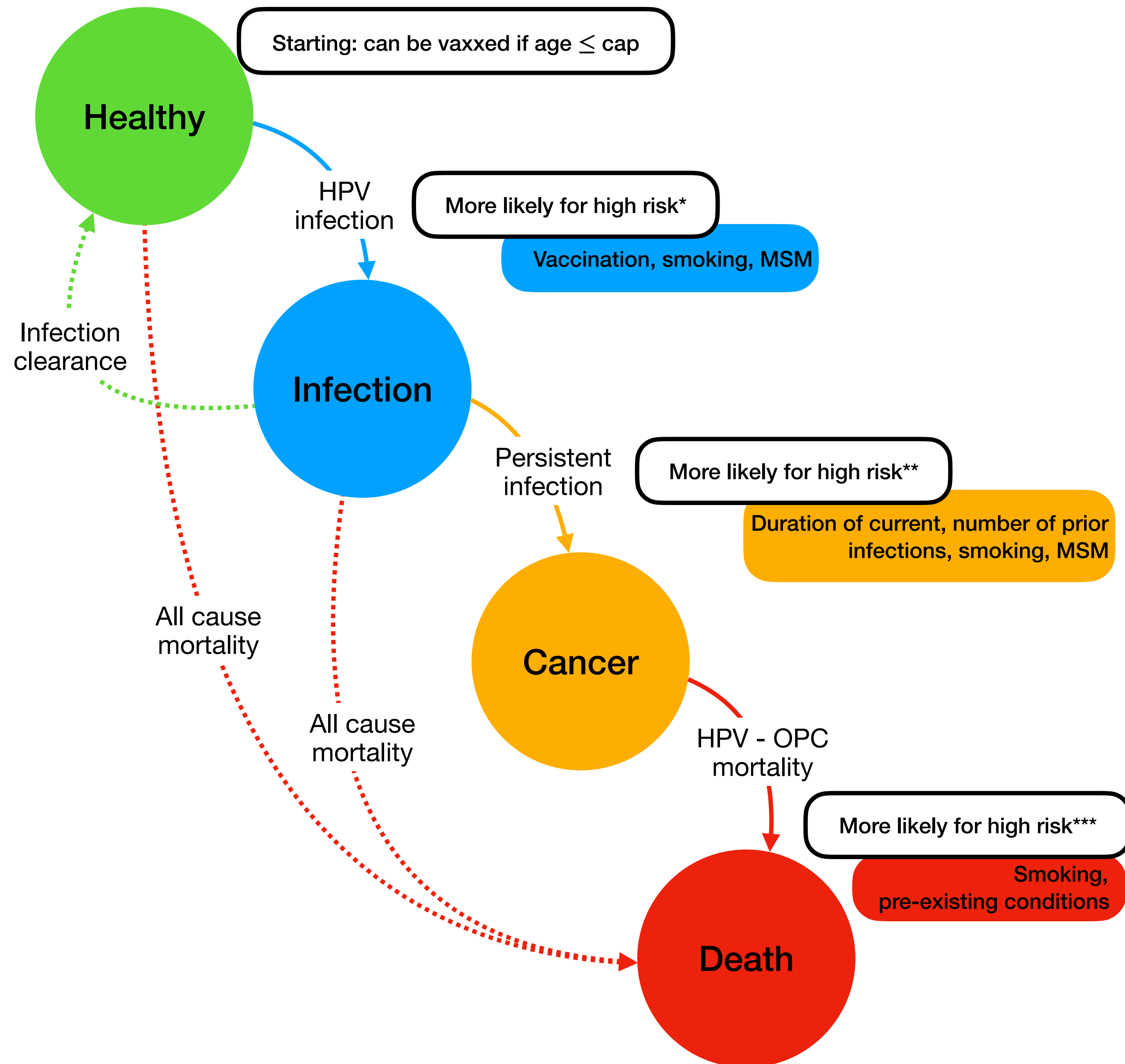


Designing an ABM Transition diagram

Generate starting population and run simulation for 60 cycles.
For a fixed cycle: iterate through EACH living person in population.

Iteration and updating rule:

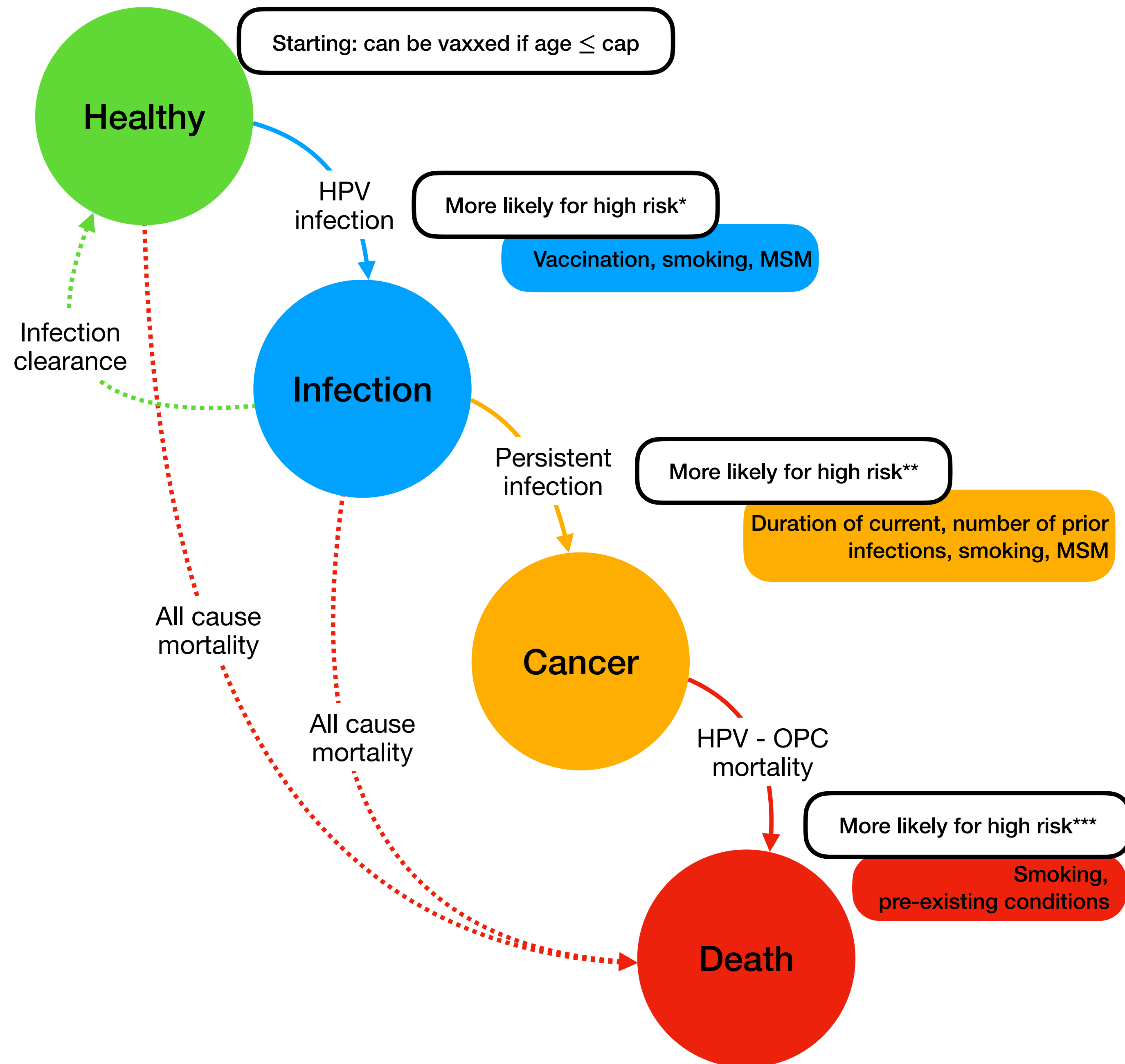
1. If **healthy**:
 1. If **unvaccinated**:
 1. Simulate vaccination (under age cap), go to next year
 2. Simulate **infection** (high risk), go to next year
 3. Simulate **death** (all-cause mortality), leave population
 2. If **vaccinated**:
 1. Simulate **infection** (low risk), go to next year
 2. Simulate **death** (all-cause mortality), leave population
2. If **infected**:
 1. Simulate return to **healthy**: infection clearance, go to next year
 2. Simulate **infection**, go to next year
 3. Simulate **cancer** (high risk), go to next year
 4. Simulate **death** (all-cause mortality), leave population
3. If **cancer**:
 1. **Cancer** persists, go to next year.
 2. Simulate **death** (high risk): OPC mortality, leave population



Designing an ABM Transition diagram

What we have done so far:

1. **Cost Analysis:** Tracks and discounts (@3.00%)
 1. Vaccination costs
 2. OPC treatment costs
 3. Long-term care costs
2. **Scenario Comparison:** Runs simulations for different age caps
 1. Cap values: 26, 30, 35, 40, and 45
3. **Visualization:** Generates plots for:
 1. Cumulative OPC cases by age cap
 2. Cumulative OPC deaths by age cap
 3. Cumulative costs by age cap



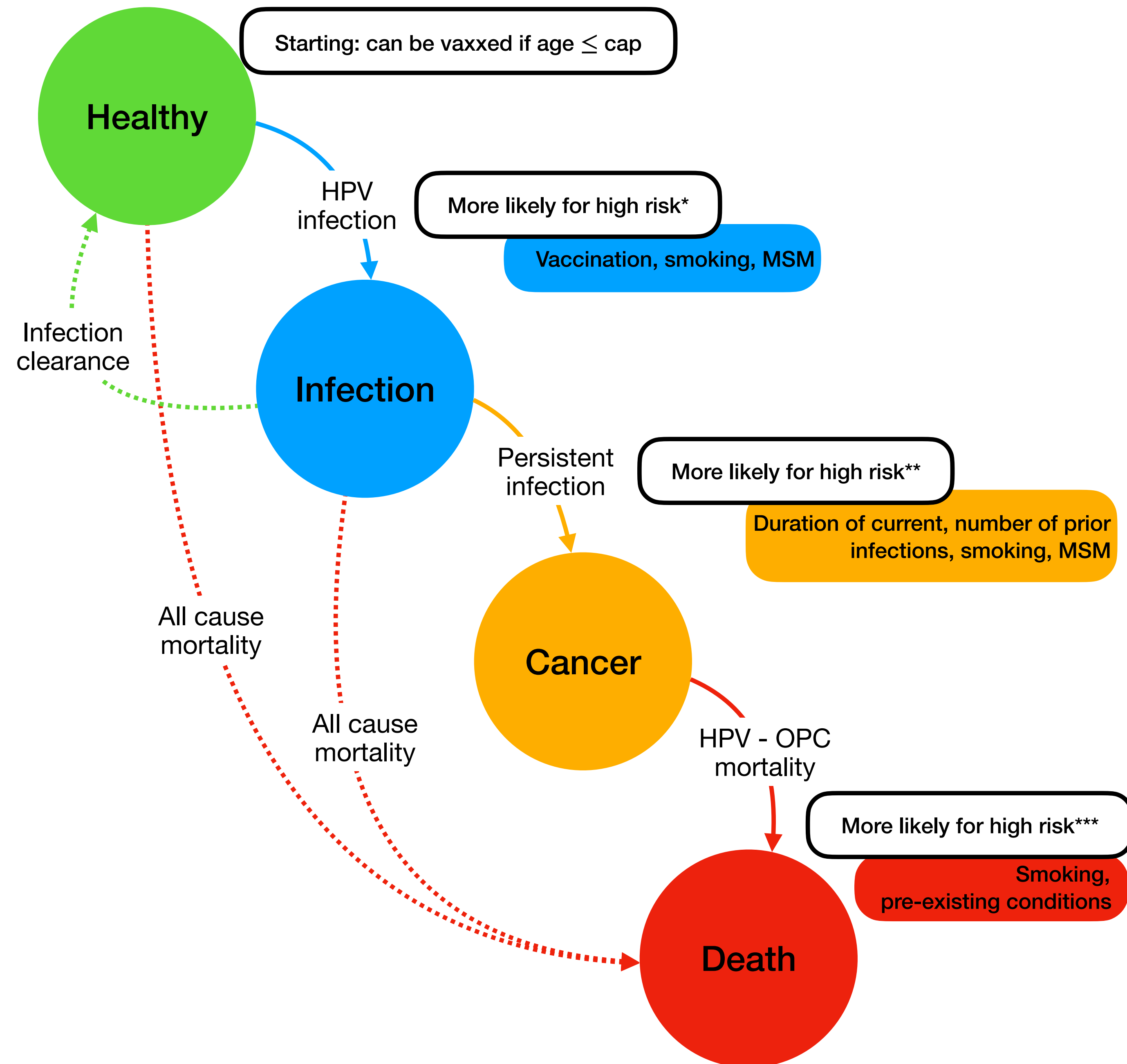
Designing an ABM Transition diagram

Next steps: calibrate this model with real VHA data for:

1. Age distribution of Veterans
2. Smoking prevalence
3. Vaccination uptake rates by age*
4. HPV prevalence*
5. Mortality rates*
6. Actual costs of vaccination and OPC treatment (easy to get)

* may have to use national data and use VA-adjustment factor

* Sensitivity analyses may be needed.



Designing an ABM

Findings from simulation study (current)

